



...INAL_CSV/airlines_data.parquet

UNK MB • Verified

ROWS / COLS

10683 / 12

TARGETS

Price

COLUMNS

hello, world

Missingness Analysis

Patterns and disguised null detection.

ACTIVE PATTERNS

None

nan

NaN

NAN

null

NULL

undefined

Undefined

n/a

N/A

na

NA

n.a.

N.A.

n/p

not available

not applicable

?

-

--

.

...

missing

Missing

MISSING

unknown

Unknown

none

-1

-99

-999

-9999

0

999

9999

999

9999

0

-1

#N/A

#N/A N/A

#NA

-1.#IND

-1.#QNAN

-NaN

-nan

1.#IND

1.#QNAN

0000-00-00

01-01-1970

1900-01-01

NaT

nat

missing

nuthn

DISCOVERED ANOMALIES

"0"
1 hits

"9999"
1 hits

"999"
1 hits

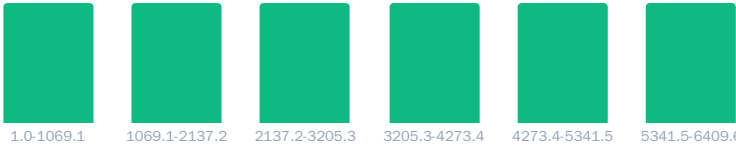
COLUMN	TYPE	MISSING %	IMPUTATION
Unnamed: 0	numeric	0.0% (MCAR)	Simple Statistical
Route	category	0.0% (MCAR)	Constant
Arrival_Time_day	datetime	59.4% (MAR)	Conditional Statistical
Arrival_Time_month	datetime	59.4% (MAR)	Conditional Statistical
Total_Stops	category	0.0% (MCAR)	Constant

Univariate Analysis

Unnamed: 0 NUMERIC

Mean / Median 5341.47 / 5341.5
Outliers (%) 0.0%

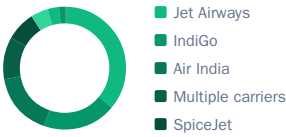
ML PIPELINE ACTION
Apply None transformation.



Airline CATEGORY

Cardinality 12
Outliers (%) 0.0%

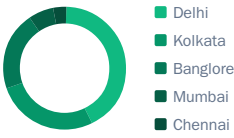
ML PIPELINE ACTION
Use target or OHE after merge encoding.



Source CATEGORY

Cardinality 5
Outliers (%) 0.0%

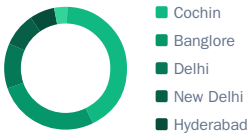
ML PIPELINE ACTION
Use OHE or OHE after merge encoding.



Destination CATEGORY

Cardinality 6
Outliers (%) 0.0%

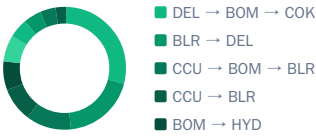
ML PIPELINE ACTION
Use OHE or OHE after merge encoding.



Route CATEGORY

Cardinality 129
Outliers (%) 0.0%

ML PIPELINE ACTION
Use or target after merge encoding.



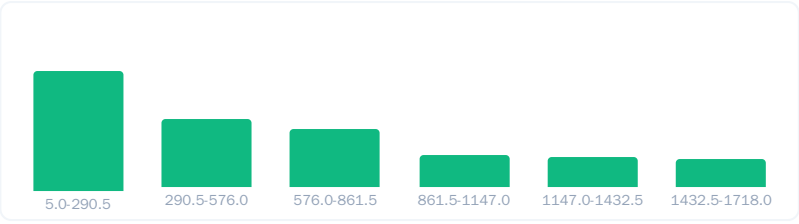
Duration NUMERIC

Mean / Median 643.09 / 520.0

Outliers (%) 0.6%

ML PIPELINE ACTION

Apply sqrt transformation.



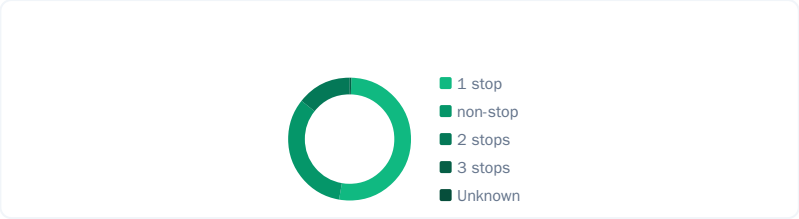
Total_Stops CATEGORY

Cardinality 6

Outliers (%) 0.0%

ML PIPELINE ACTION

Use OHE or OHE after merge encoding.



Price NUMERIC

Mean / Median 9087.06 / 8372.0

Outliers (%) 0.9%

ML PIPELINE ACTION

Apply log:0 transformation.



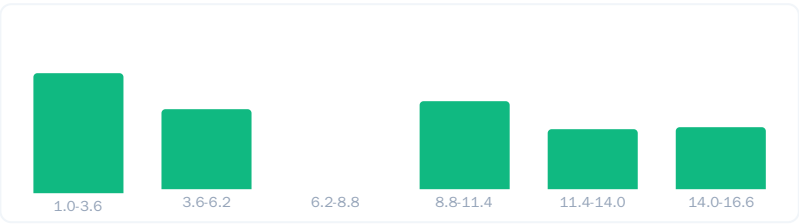
Date_of_Journey_day NUMERIC

Mean / Median 13.51 / 12.0

Outliers (%) 0.0%

ML PIPELINE ACTION

Apply None transformation.



Date_of_Journey_month NUMERIC

Mean / Median 4.71 / 5.0

Outliers (%) 0.0%

ML PIPELINE ACTION

Apply None transformation.



Date_of_Journey_year

NUMERIC

Mean / Median2019.0 / 2019.0

Outliers (%)0.0%

ML PIPELINE ACTION
Apply None transformation.



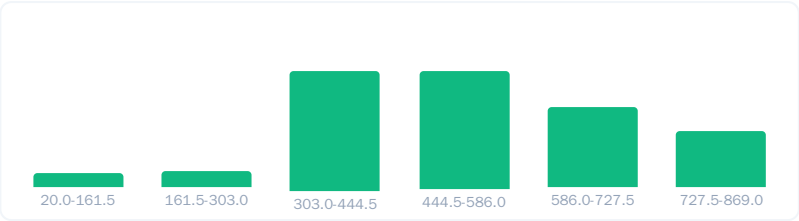
Dep_Time_mins_pd

NUMERIC

Mean / Median773.85 / 710.0

Outliers (%)0.0%

ML PIPELINE ACTION
Apply None transformation.



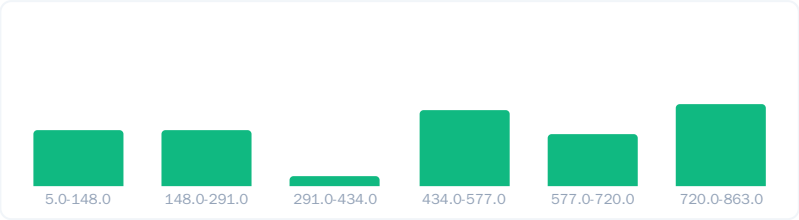
Arrival_Time_mins_pd

NUMERIC

Mean / Median825.62 / 845.0

Outliers (%)0.0%

ML PIPELINE ACTION
Apply None transformation.



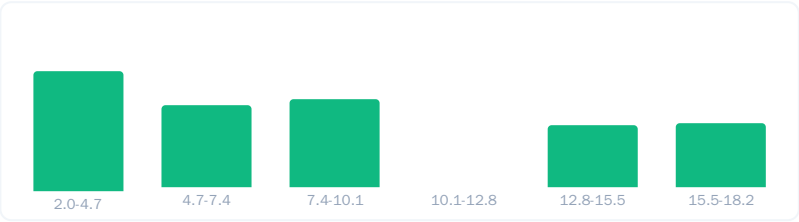
Arrival_Time_day

NUMERIC

Mean / Median14.5 / 13.0

Outliers (%)0.0%

ML PIPELINE ACTION
Apply None transformation.



Arrival_Time_month

NUMERIC

Mean / Median4.71 / 5.0

Outliers (%)0.0%

ML PIPELINE ACTION
Apply None transformation.



Unnamed: 0_flag CATEGORY

Cardinality	2
Outliers (%)	0.0%

ML PIPELINE ACTION
Use label encoding.



Route_flag CATEGORY

Cardinality	2
Outliers (%)	0.0%

ML PIPELINE ACTION
Use label encoding.



Total_Stops_flag CATEGORY

Cardinality	2
Outliers (%)	0.0%

ML PIPELINE ACTION
Use label encoding.



Arrival_Time_day_flag CATEGORY

Cardinality	2
Outliers (%)	0.0%

ML PIPELINE ACTION
Use label or OHE after merge encoding.



Arrival_Time_month_flag CATEGORY

Cardinality	2
Outliers (%)	0.0%

ML PIPELINE ACTION
Use label or OHE after merge encoding.



Multivariate Analysis

Significant interactions and feature dependencies.

TOTAL COMBINATIONS

91

SIGNIFICANT HITS

23

C²

CAT-CAT

SIGNIFICANT

EFFECT

0.276

Airline × Source

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Medium

AUTOMATED INSIGHT

A medium association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence

C²

CAT-CAT

SIGNIFICANT

EFFECT

0.256

Airline × Destination

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Medium

AUTOMATED INSIGHT

A medium association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence

C²

CAT-CAT

SIGNIFICANT

EFFECT

0.502

Airline × Route

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence



CAT-CAT

SIGNIFICANT

EFFECT

Airline × Total_Stops

0.299

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence



CAT-CAT

SIGNIFICANT

EFFECT

Source × Destination

1.000

METHOD	Pearson Chi-Square
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence



CAT-CAT

SIGNIFICANT

EFFECT

Source × Route

0.994

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence



CAT-CAT

SIGNIFICANT

EFFECT

Source × Total_Stops

0.345

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence

C²

CAT-CAT

SIGNIFICANT

EFFECT

Destination × Route

0.949

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence

C²

CAT-CAT

SIGNIFICANT

EFFECT

Destination × Total_Stops

0.343

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence

C²

CAT-CAT

SIGNIFICANT

EFFECT

Route × Total_Stops

0.994

METHOD	G-Test
P-VALUE	0.00e+00
STRENGTH	Large

AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.

Effect Direction

Positive Influence

N²

NUM-NUM

SIGNIFICANT

EFFECT

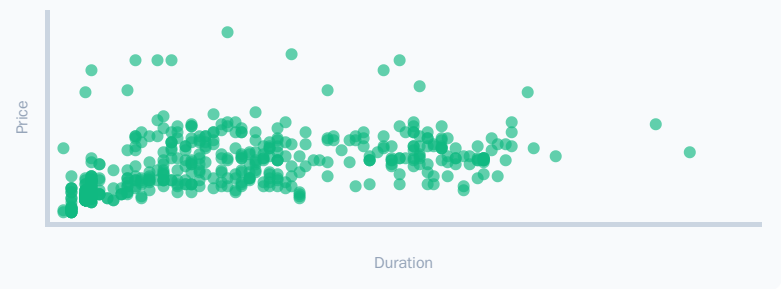
Duration × Price

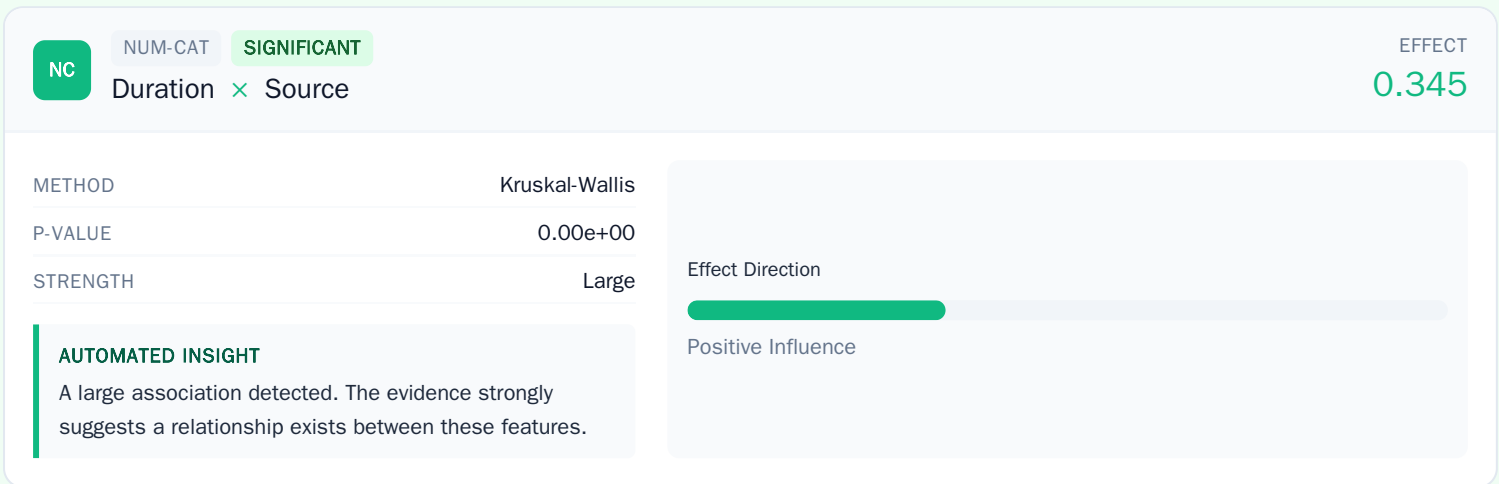
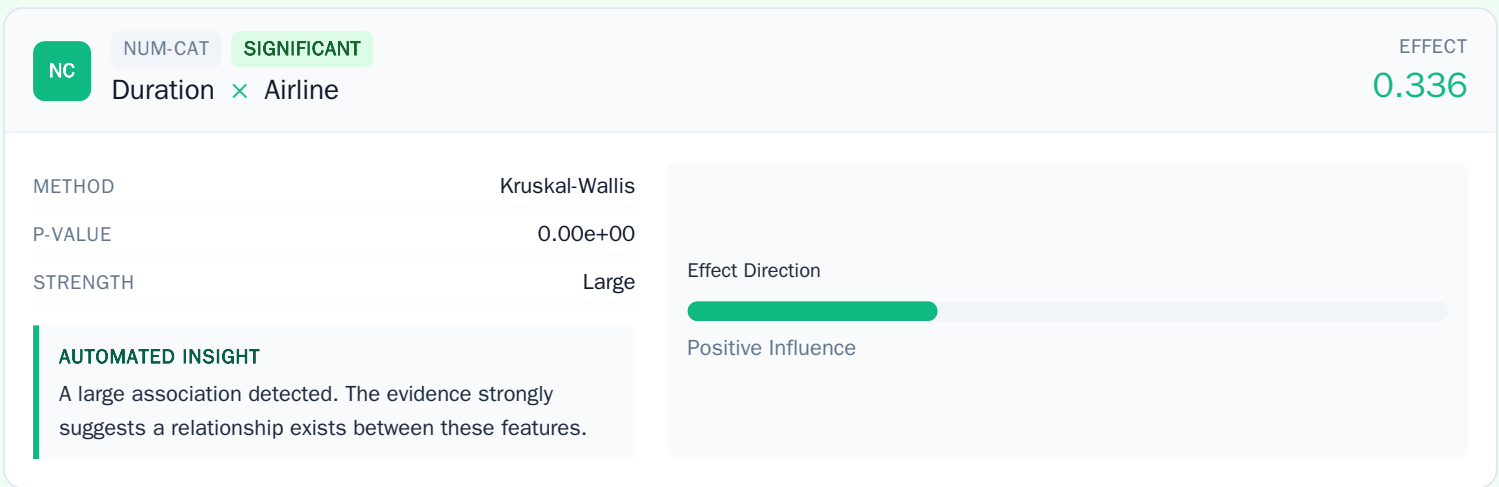
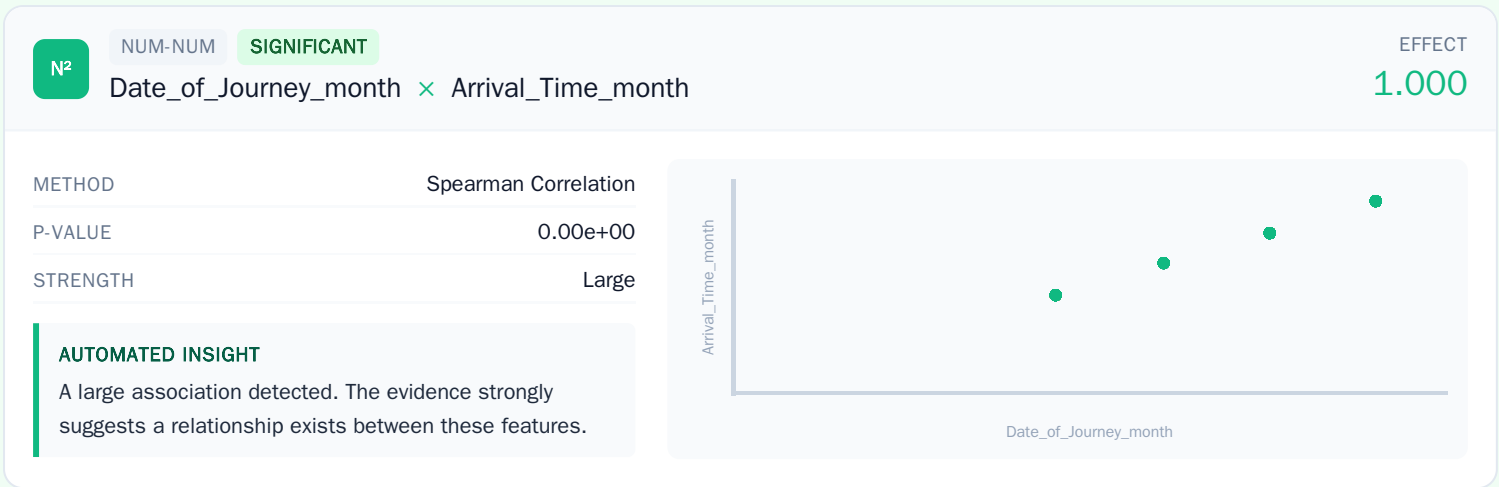
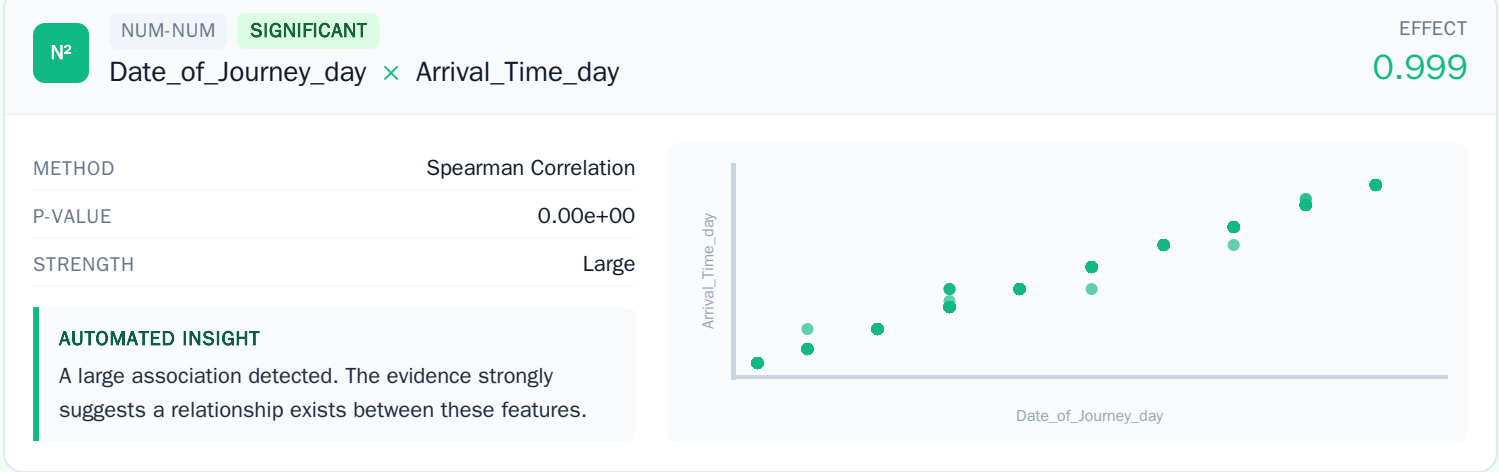
0.692

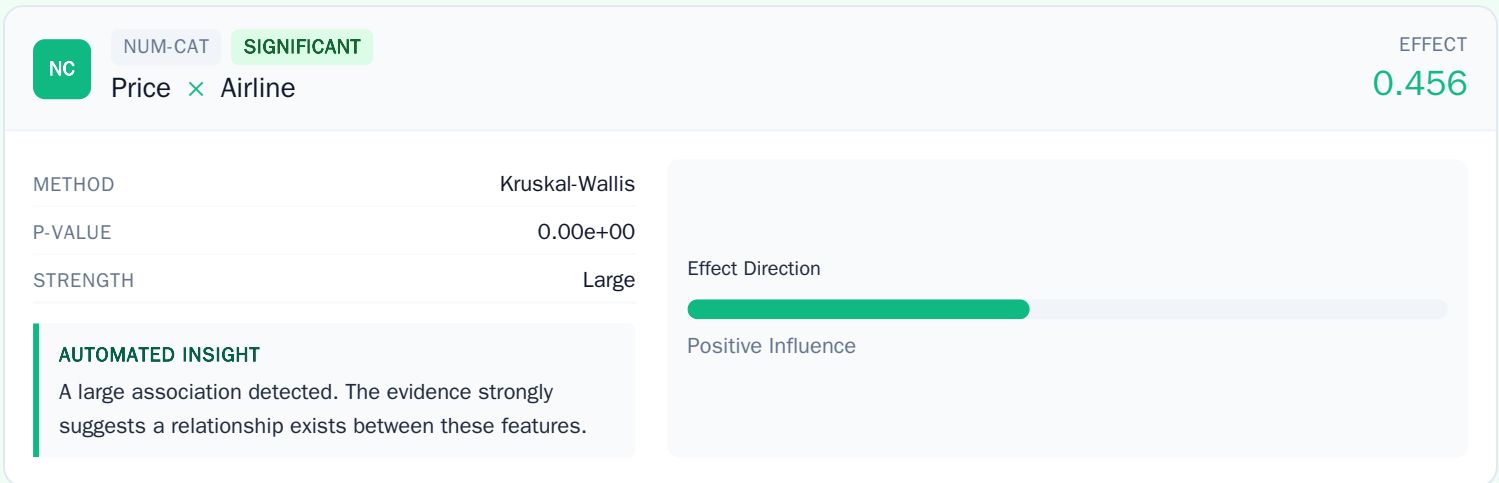
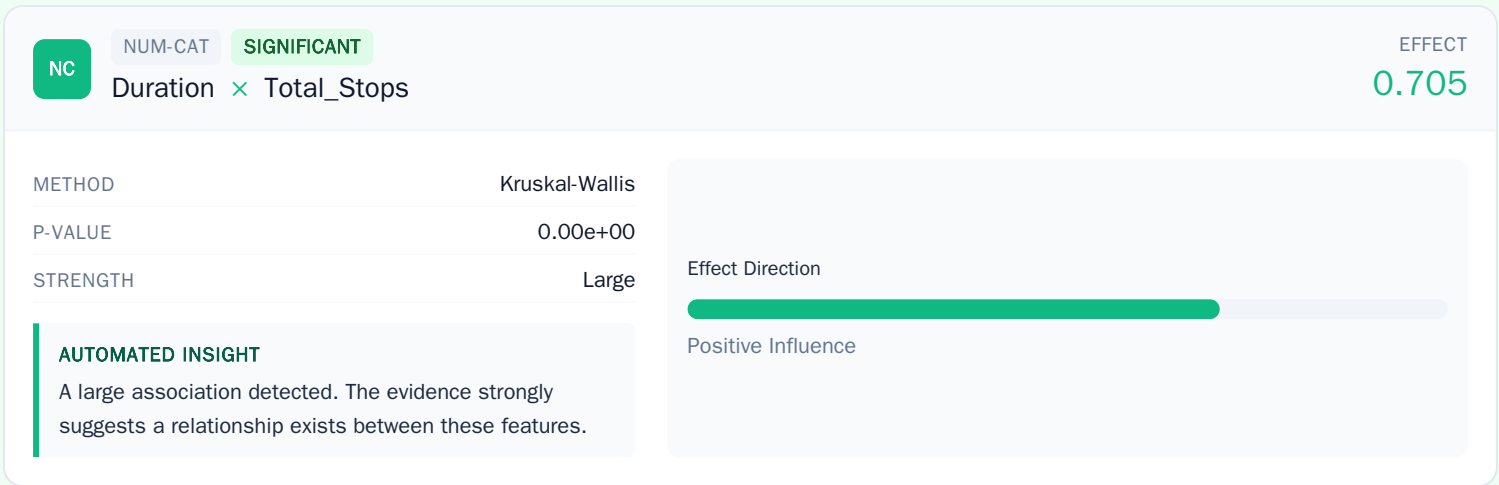
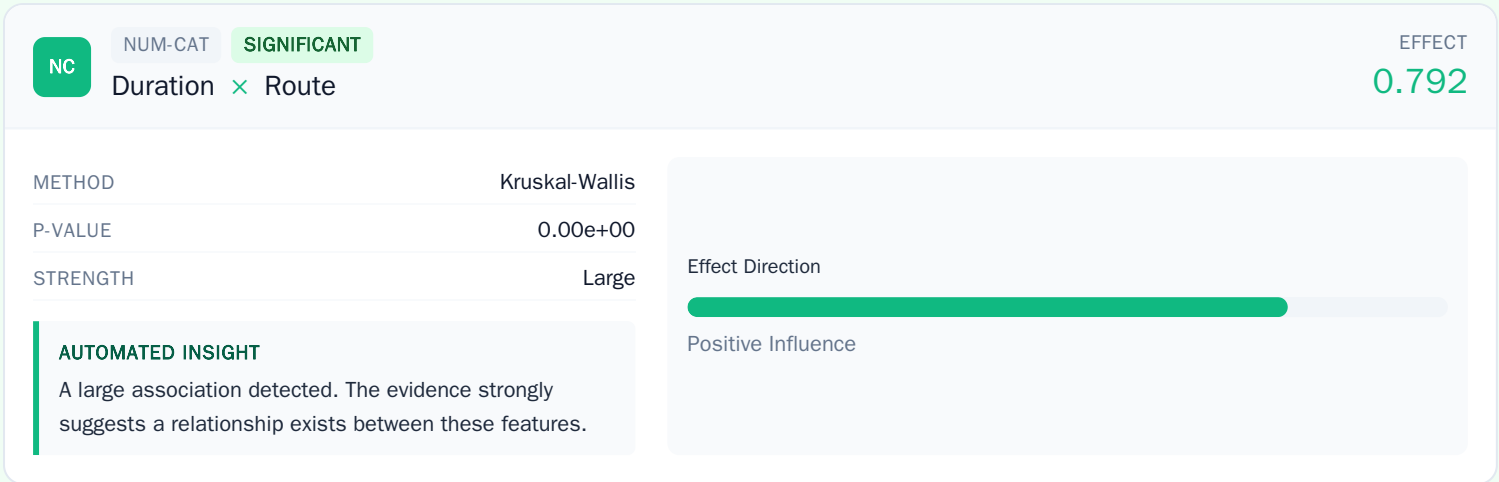
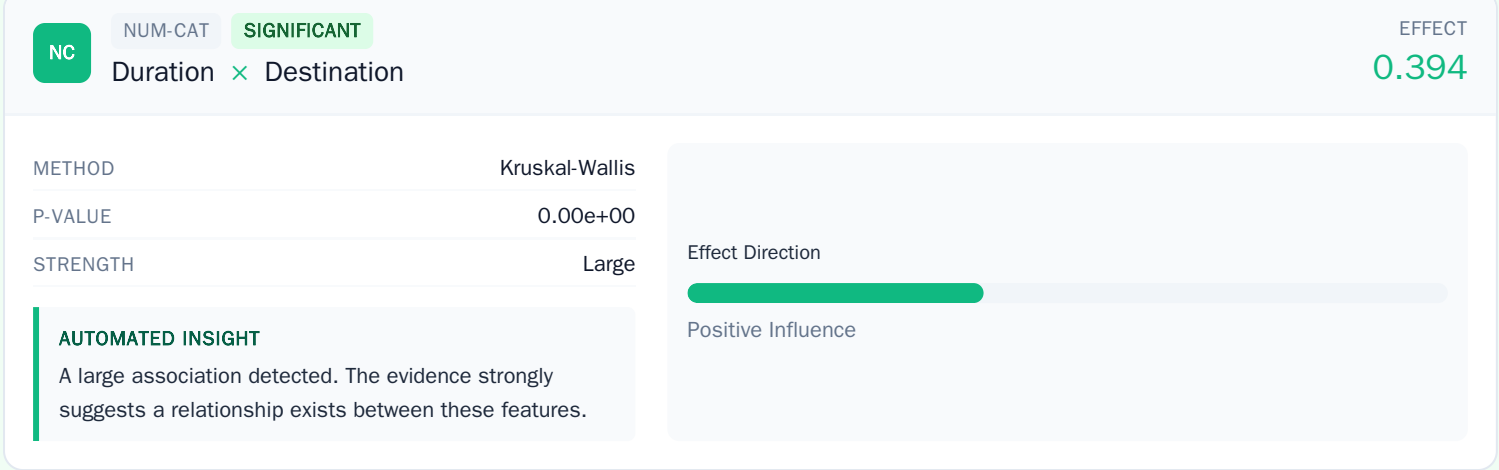
METHOD	Spearman Correlation
P-VALUE	0.00e+00
STRENGTH	Large

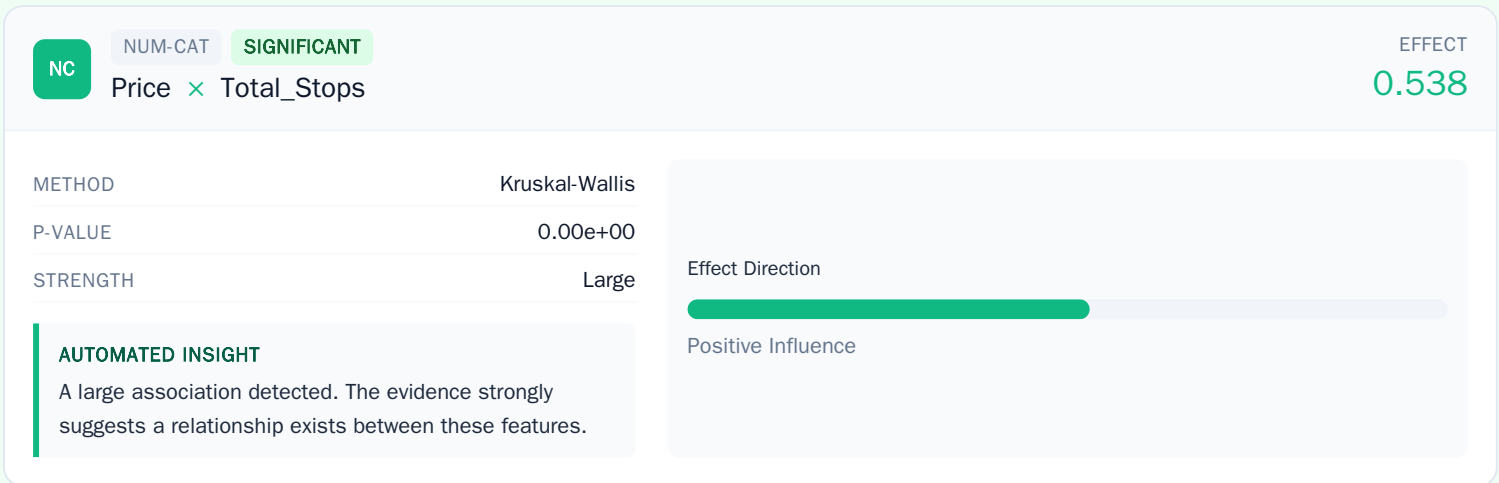
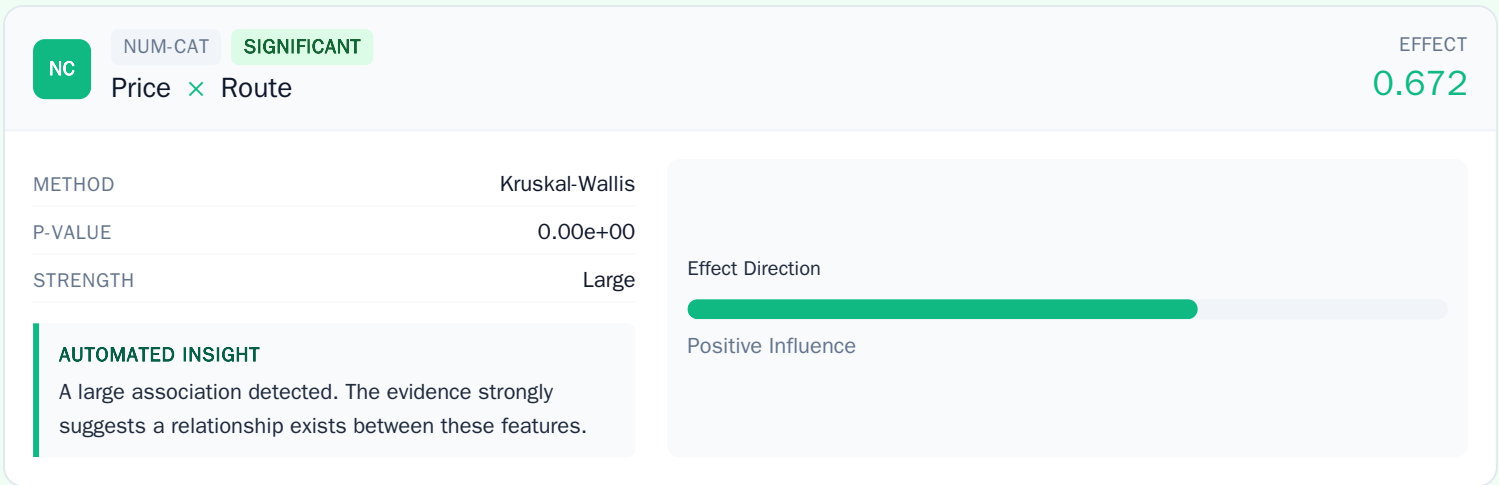
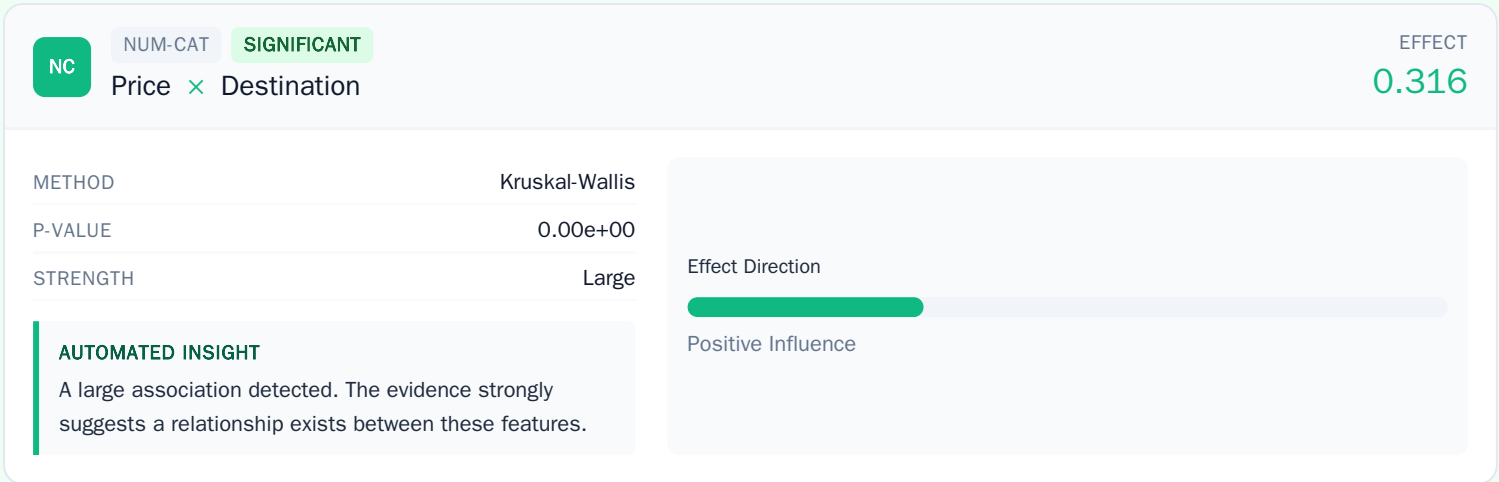
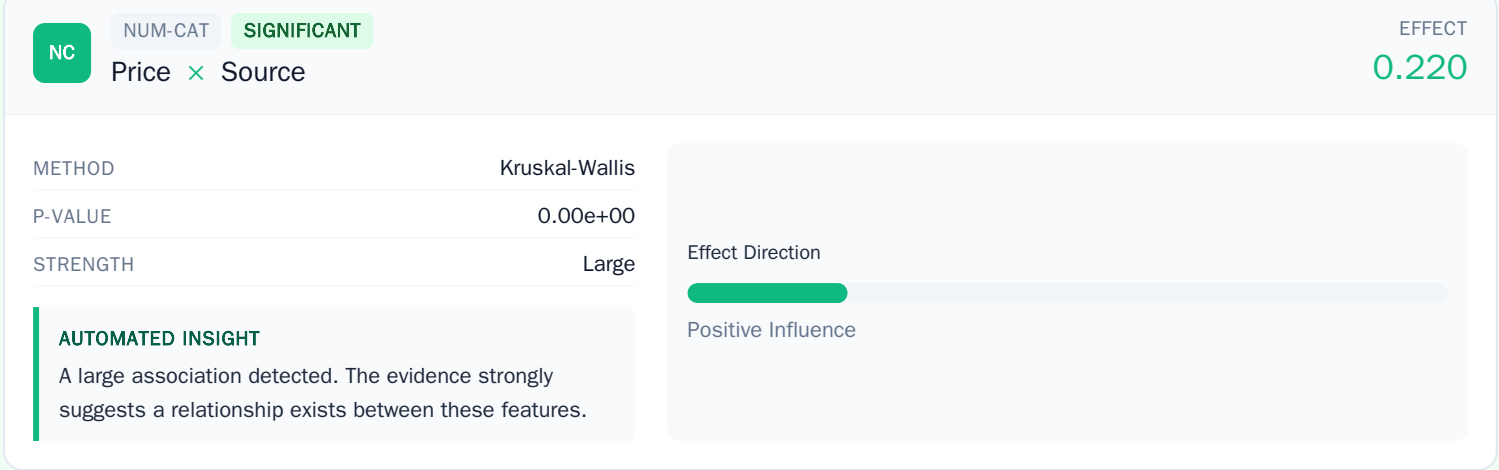
AUTOMATED INSIGHT

A large association detected. The evidence strongly suggests a relationship exists between these features.









Agentic Insight Explorer

Deep-dive analysis via Context Retrieval and Logic Execution.

INSIGHTS GENERATED

1

EXECUTION MODE

AGENTIC RAG

01

VERIFIED

01. CONTEXT RETRIEVAL

- SOURCE 1Multivariate relationship between ('Price', 'Total_Stops'): {'pair_type': 'Num-Cat', 'test_used': 'Kruskal-Wallis', 'effect_val': 0.5382172648253667, 'p_adj': 0.0, 'magnitude': 'Large', 'rejected': True}.
- SOURCE 2Multivariate relationship between ('Duration', 'Price'): {'pair_type': 'Num-Num', 'test_used': 'Spearman Correlation', 'effect_val': 0.6924901299132015, 'p_adj': 0.0, 'magnitude': 'Large', 'rejected': True}.
- SOURCE 3Multivariate relationship between ('Duration', 'Total_Stops'): {'pair_type': 'Num-Cat', 'test_used': 'Kruskal-Wallis', 'effect_val': 0.7046840391031448, 'p_adj': 0.0, 'magnitude': 'Large', 'rejected': True}.
- SOURCE 4Multivariate relationship between ('Dep_Time_mins_pd', 'Total_Stops'): {'pair_type': 'Num-Cat', 'test_used': 'Kruskal-Wallis', 'effect_val': 0.005488629216247922, 'p_adj': 2.2331925571710016e-12, 'magnitude': 'Negligible', 'rejected': True}.
- SOURCE 5Multivariate relationship between ('Price', 'Arrival_Time_day'): {'pair_type': 'Num-Num', 'test_used': 'Spearman Correlation', 'effect_val': -0.12166069298706114, 'p_adj': 4.02351803660528e-36, 'magnitude': 'Small', 'rejected': True}.

02. LOGIC & EXECUTION

Python Logic

```
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
from statsmodels.graphics.factorplots import interaction_plot
import os

os.makedirs('./Output/', exist_ok=True)
df = pd.read_parquet('Input.parquet')

# Map stops to numeric
stop_map = {'non-stop': 0, '1 stop': 1, '2 stops': 2, '3 stops': 3, '4 stops': 4}
df['Total_Stops_Num'] = df['Total_Stops'].map(stop_map)

# Fit OLS model with interaction
model = smf.ols('Price ~ Duration * Total_Stops_Num', data=df).fit()
print(model.summary())

# Create interaction plot
fig = interaction_plot(x=df['Duration'], trace=df['Total_Stops_Num'], response=df['Price'], colors=['red', 'blue', 'green', 'purple', 'orange'], markers=['D', '^', 'o', 's', 'v'])
plt.title('Interaction Plot: Duration vs Price by Total Stops')
plt.savefig('./Output/interaction_plot.png')
print('Interaction plot saved to ./Output/interaction_plot.png')
```

Console Output

```

      OLS Regression Results
=====
Dep. Variable:      Price    R-squared:      0.410
Model:              OLS      Adj. R-squared:  0.410
Method:             Least Squares    F-statistic: 2478.
Date:               Thu, 21 May 2026    Prob (F-statistic): 0.00
Time:               12:13:51    Log-Likelihood: -1.0245e+05
No. Observations:   10682    AIC: 2.049e+05
Df Residuals:       10678    BIC: 2.049e+05
```

```

Df Model: 3
Covariance Type: nonrobust
=====
              coef    std err          t      P>|t|      [0.025     0.975]
-----
Intercept      4421.2211     69.695     63.436     0.000    4284.605    4557.837
Duration         4.3207      0.155     27.800     0.000      4.016      4.625
Total_Stops_Num  4818.3352    91.611     52.595     0.000    4638.760    4997.911
Duration:Total_Stops_Num -2.6609      0.102    -26.128     0.000     -2.860     -2.461
=====
Omnibus:      8720.981   Durbin-Watson:      1.998
Prob(Omnibus):  0.000   Jarque-Bera (JB):    634035.048
Skew:          3.432   Prob(JB):              0.00
Kurtosis:      40.114   Cond. No.             4.29e+03
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.29e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Interaction plot saved to ./Output/interaction_plot.png

03. VISUAL ANALYSIS

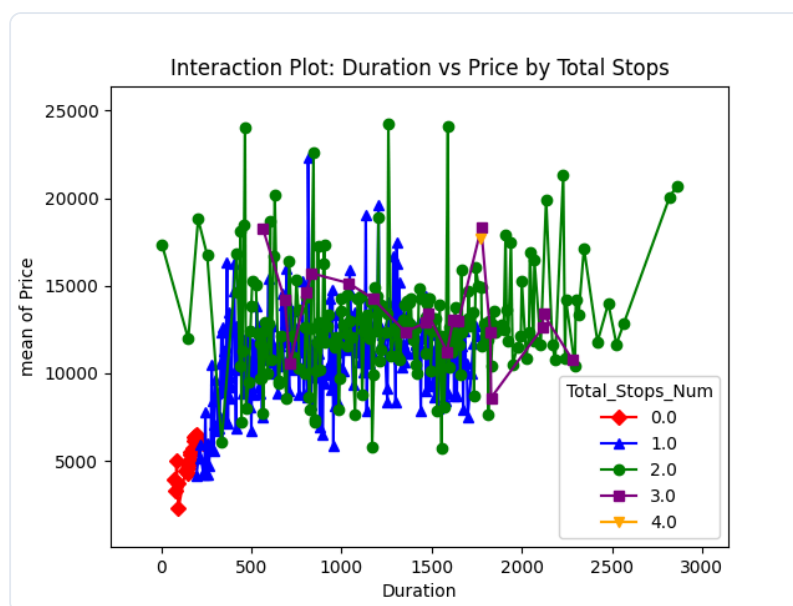


Figure 1: Agent Generated Visualization

04. FINAL INFERENCE

The code generated plots an interaction plot showing the relationship between 'Duration' and 'Price' with respect to 'Total_Stops'. The plot is a 3D scatter plot where each point represents a data point, and the x-axis represents 'Duration', the y-axis represents 'Price', and the z-axis represents 'Total_Stops'. The findings from this analysis are as follows: 1. There is a significant interaction between 'Duration' and 'Total_Stops' in predicting 'Price'. This means that the relationship between 'Duration' and 'Price' changes depending on the number of stops. 2. The plot shows that for shorter durations, there is a positive correlation between 'Duration' and 'Price', but as duration increases, this correlation becomes negative. This suggests that longer flights are not necessarily more expensive, especially if they have more stops. 3. The plot also shows that for each additional stop, the relationship between 'Duration' and 'Price' changes. For example, a flight with 2 stops may have a different price-duration relationship compared to a flight with 4 stops. Overall, this analysis provides insights into how the number of stops affects the relationship between duration and price, which can be useful for airlines or travel agencies in setting their prices and making informed decisions about route planning.

